MSHA Narrative Clustering Proof of Concept

Updated Vocabulary and Nine-Cluster Analysis

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Narratives analyzed: 2,000
Vocabulary terms retained: 2,763
Clusters used: 9
Nine-cluster silhouette score: 0.0533
Highest tested silhouette score: 0.0736
Highest-scoring tested cluster count: 15
Latent semantic dimensions: 75
Explained variance: 0.227

Purpose:
Test whether meaningful narrative groups can be discovered using
terms and phrases that are prevalent but not overly common.

Interpretation:
Clusters are exploratory text-similarity groups. They are not
verified causes, official classifications, or final safety topics.
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Executive Summary

This proof-of-concept analysis grouped 2,000 MSHA accident narratives using only the narrative text. Existing MSHA classification, accident type, activity, injury source, and injury severity fields were not used to create the clusters.

A nine-cluster solution was retained for interpretation. Its silhouette score was 0.0533. The highest tested score was 0.0736 using 15 clusters.

The difference between the nine-cluster and highest-scoring solutions was small. The nine-cluster solution was therefore selected as a simpler exploratory representation.

The largest cluster contained 414 narratives and was labeled 'belt | conveyor | explosion'.

The cluster with the highest mean document-level silhouette was 'roof fall | long wide | entry' with a mean value of 0.221.

531 narratives had negative silhouette values, meaning those narratives were closer to another cluster than to portions of their assigned cluster.

The results should be treated as evidence that recurring vocabulary patterns exist, not as proof that the narratives form nine definitive accident categories.

How to Interpret This Report

Cluster:
A group of narratives with similar TF-IDF and latent text features.

Cluster label:
A provisional name generated from the highest-weighted meaningful terms and phrases. Generic employee references and common English words were excluded from the label.

Representative narrative:
A narrative located relatively close to its assigned cluster center.

Borderline narrative:
A narrative with a low document-level silhouette value. Borderline records may contain several topics or may fit another cluster nearly as well.

Silhouette score:
A measure of within-cluster similarity relative to separation from other clusters. Higher values indicate clearer separation. Values near zero indicate overlap, while negative values indicate questionable assignments.

Existing fields:
Official MSHA fields are shown only after clustering. They help assess whether each narrative cluster adds information or mainly recreates an existing category.

Methodology

1. A balanced sample was selected across five injury-severity groups.

2. Common abbreviations and selected equivalent phrases were normalized. Aggressive stemming and phrase destruction were avoided.

3. Standard English stop words and common MSHA narrative boilerplate were removed.

4. Terms had to occur in at least 3 narratives and could not occur in more than 70% of the sample.

5. Both individual words and two-word phrases were retained.

6. TF-IDF assigned higher weight to terms that helped distinguish one narrative from others.

7. Truncated SVD reduced the vocabulary to 75 latent text dimensions.

8. MiniBatch K-means grouped narratives in the reduced text space.

9. Several cluster counts were evaluated. Nine clusters were retained because the nine- and fifteen-cluster silhouette scores were nearly identical in the initial test, while nine clusters were easier to interpret.

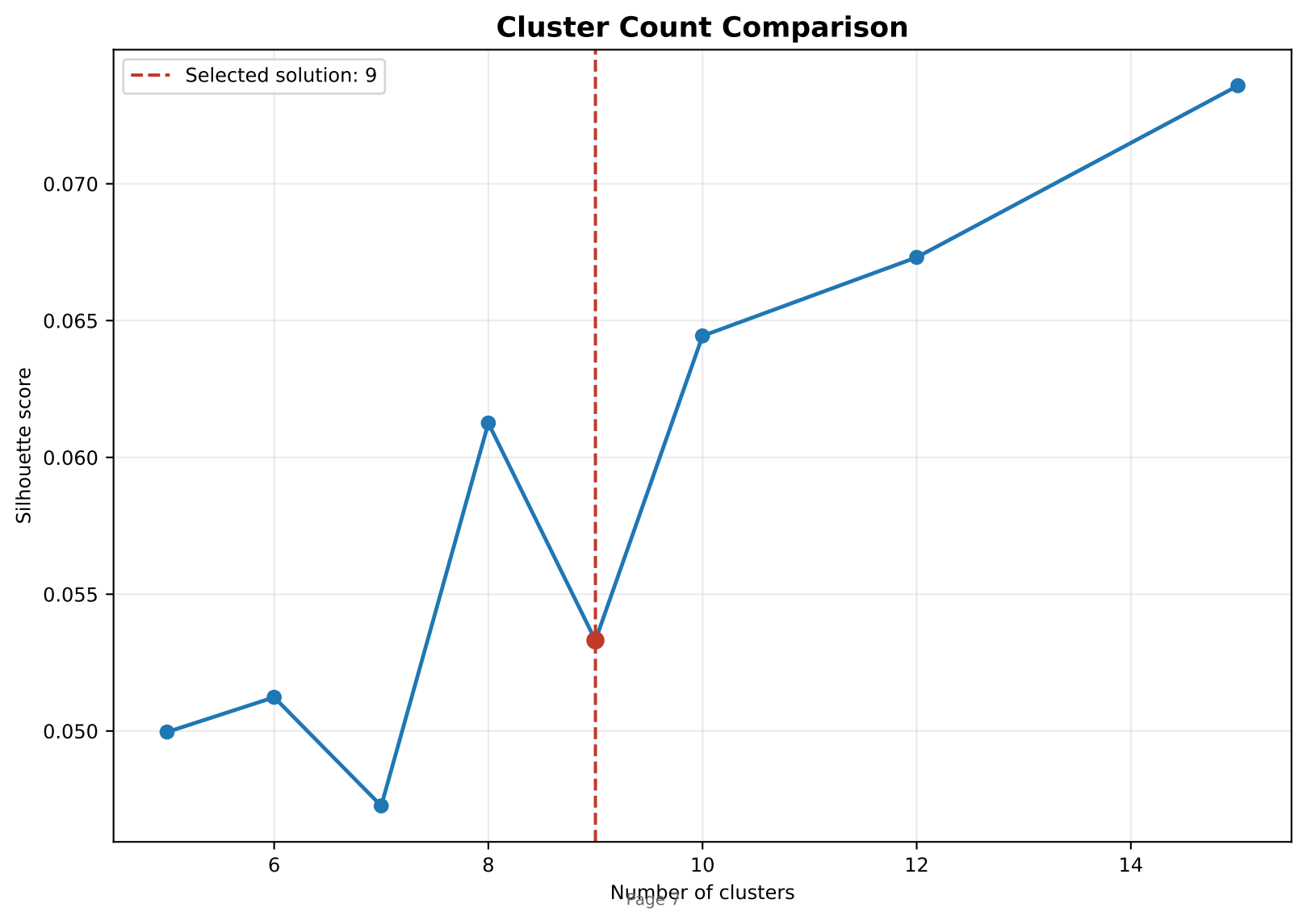
10. Existing MSHA fields were examined only after clustering.

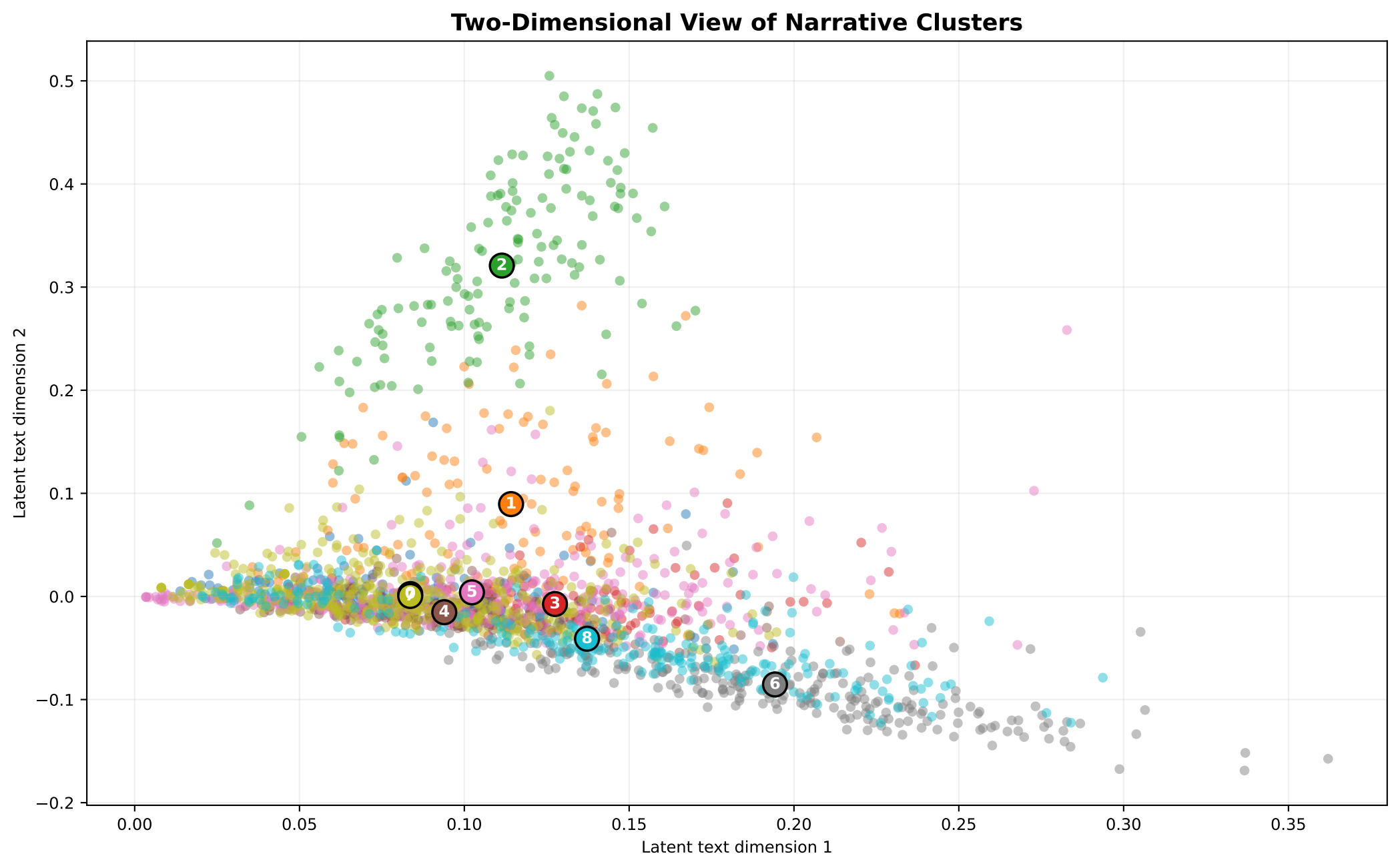
Severity-Stratified Sample

Severity Group	Available Records	Target	Selected	Final Selected
Fatality	1188	400	400	400
Permanent disability	2455	400	400	400
Days away	178432	400	400	400
Restricted or transferred	42788	400	400	400
No lost time or other	43965	400	400	400

Cluster Count Evaluation

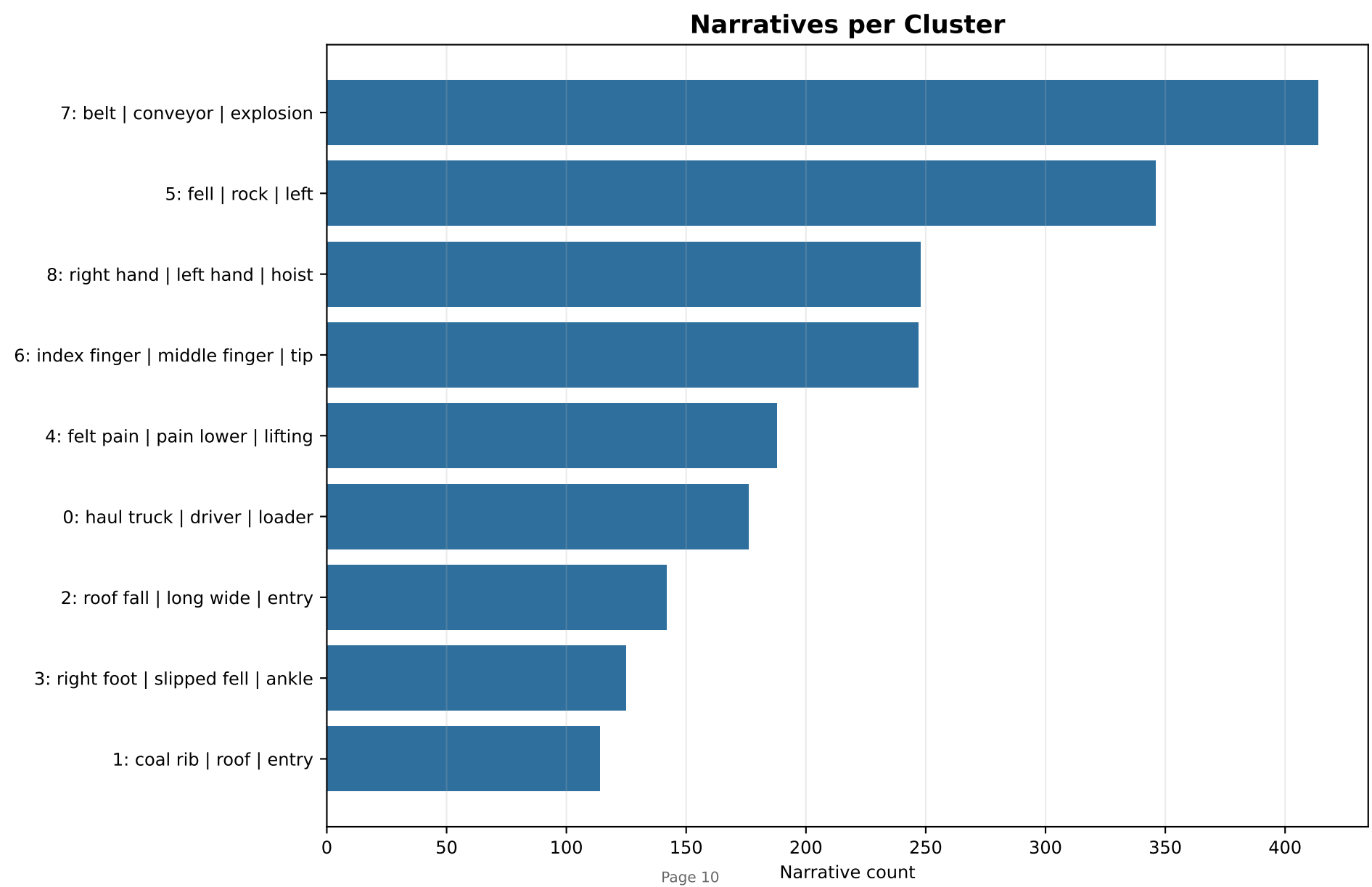
Number Of Clusters	Silhouette Score	Smallest Cluster	Largest Cluster	Median Cluster Size
5	0.0500	168	629	411.0
6	0.0512	173	481	327.0
7	0.0473	175	522	258.0
8	0.0613	111	360	262.5
9	0.0533	114	414	188.0
10	0.0644	59	334	184.0
12	0.0673	26	364	154.5
15	0.0736	55	467	105.0

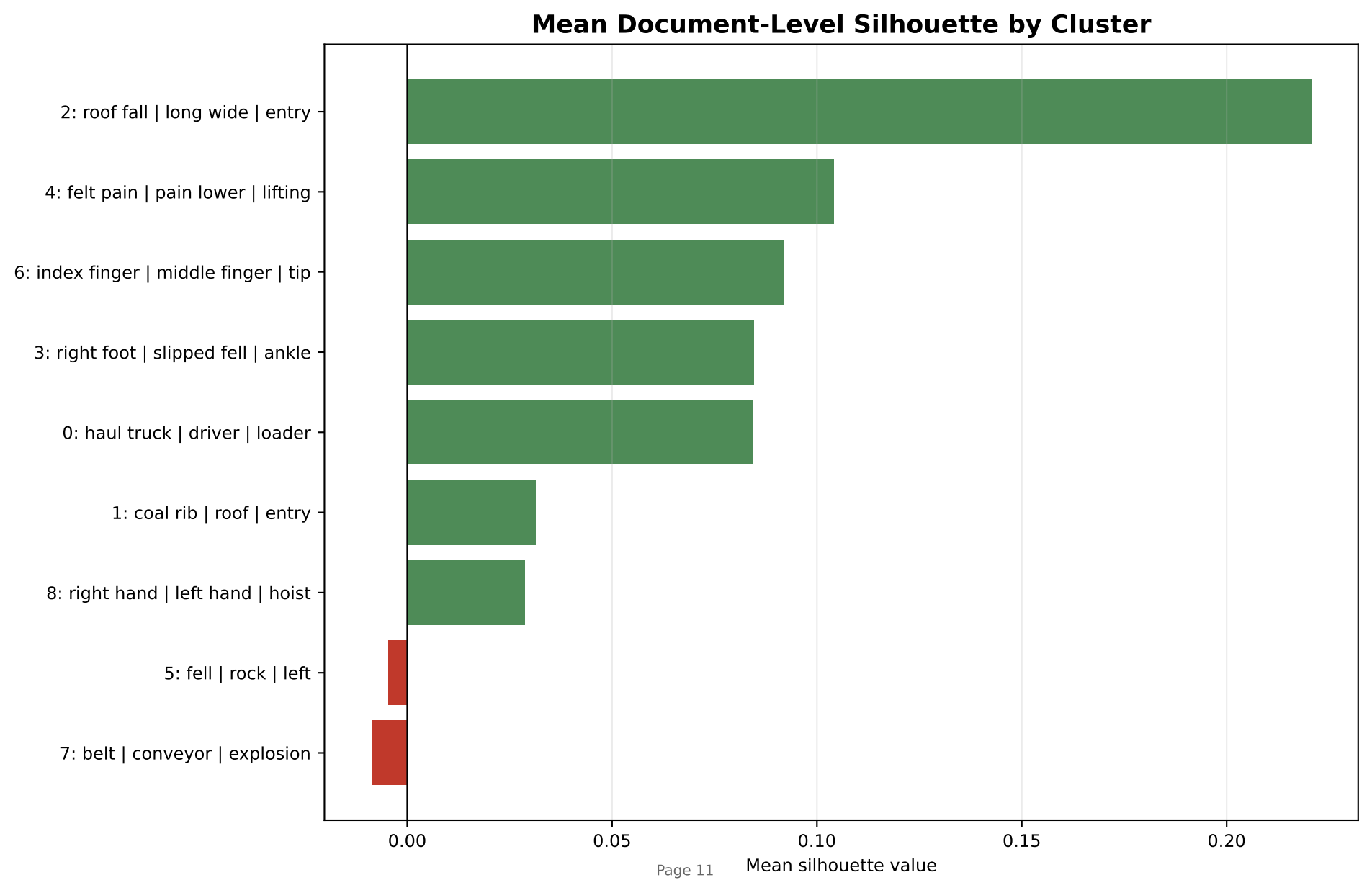

Cluster Count Comparison
Selected solution: 9
0.070
0.065
0.060
0.055
0.050
Silhouette score
6
8
10
12
14
Number of clusters

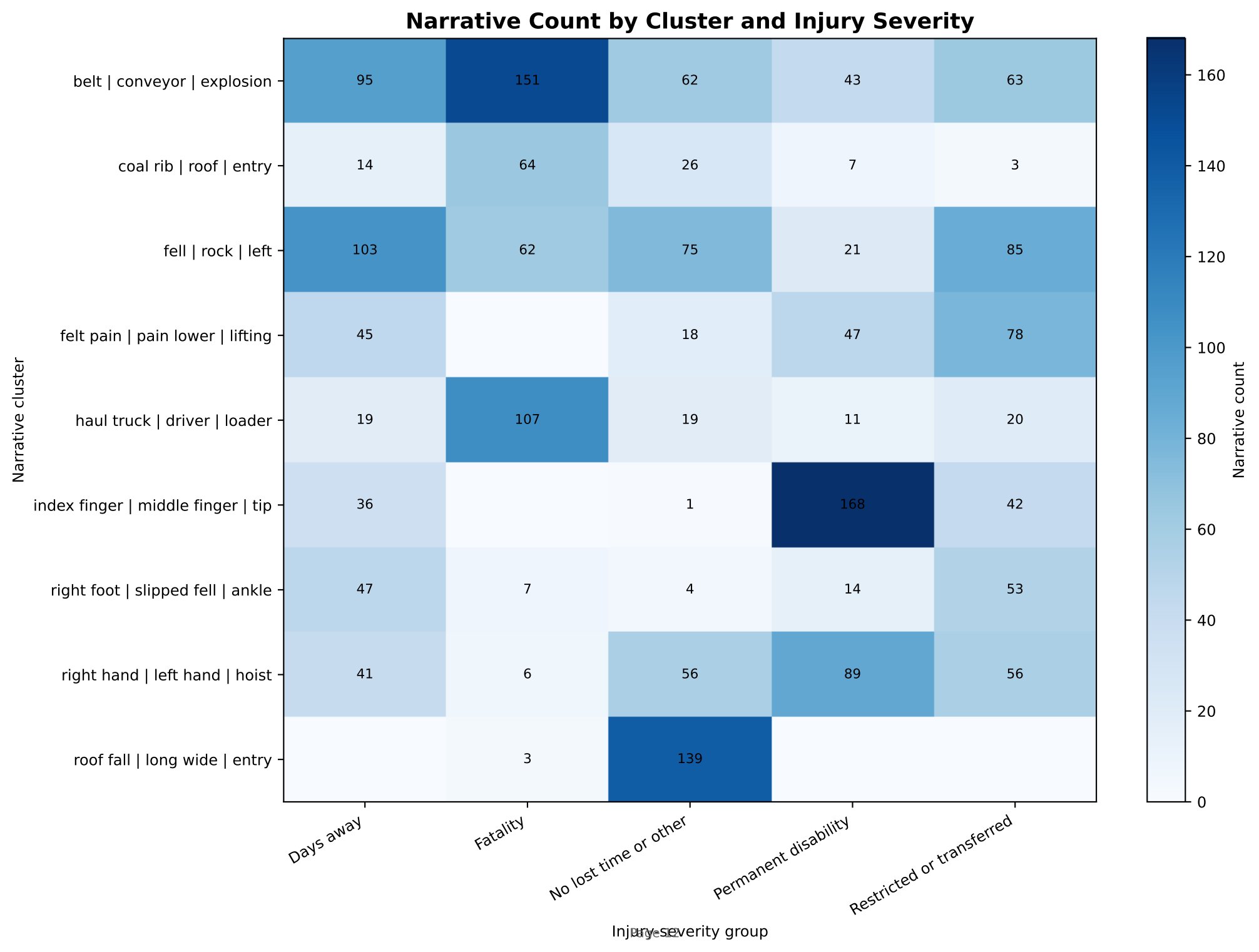

Two-Dimensional View of Narrative Clusters
Latent text dimension 1
Latent text dimension 2
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0: haul truck | driver | loader
1: coal rib | roof | entry
2: roof fall | long wide | entry
3: right foot | slipped fell | ankle
4: felt pain | pain lower | lifting
5: fell | rock | left
6: index finger | middle finger | tip
7: belt | conveyor | explosion
8: right hand | left hand | hoist

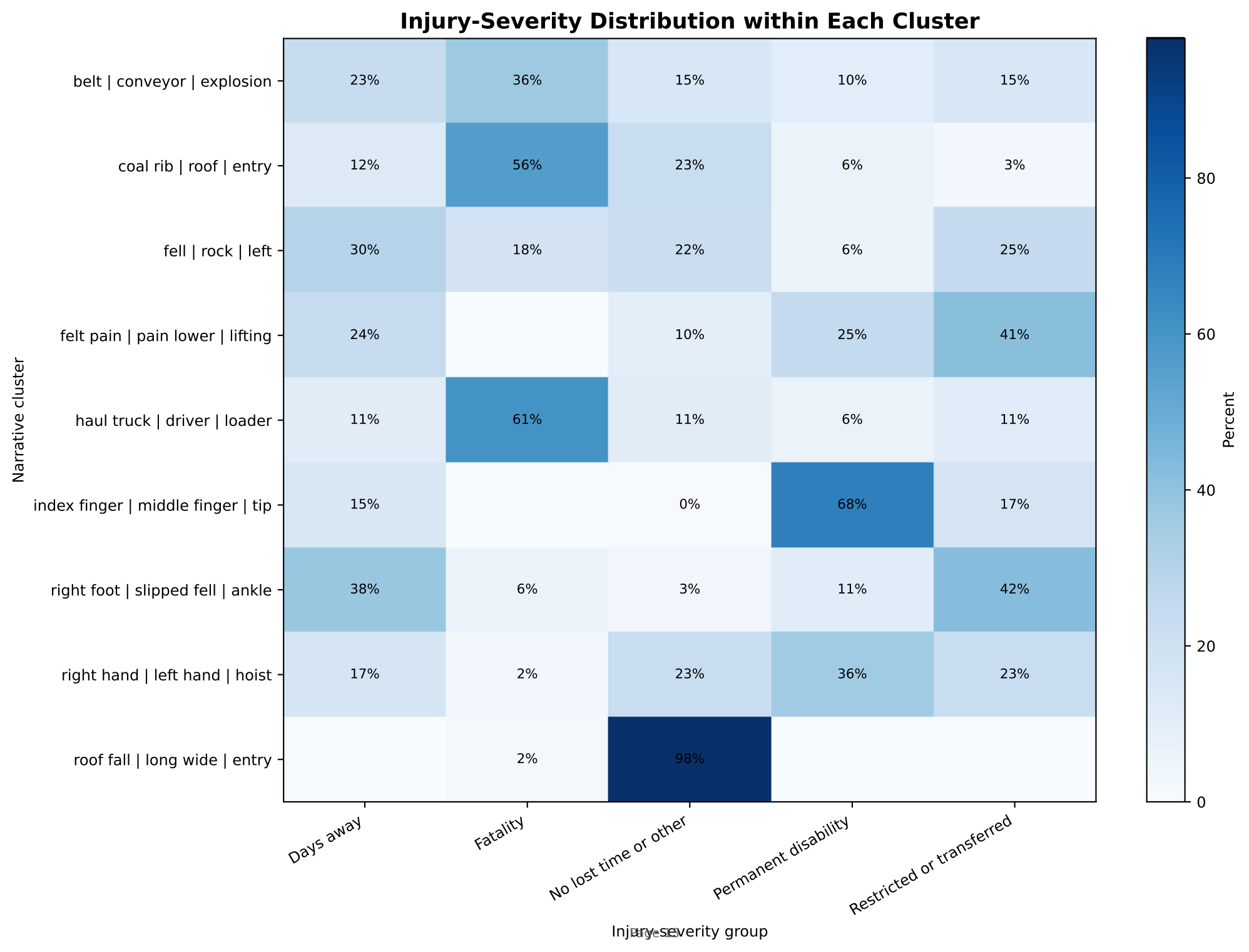
Cluster Size and Quality Summary

Cluster Id	Cluster Name	Narratives	Median Word Count	Mean Silhouette	Negative Silhouette Cc	Mean Representativen	Negative Silhouette Pct
7	belt | conveyor | explosio	414	36.0	-0.009	261	0.510	63.0%
5	fell | rock | left	346	29.0	-0.005	190	0.513	54.9%
8	right hand | left hand | hoist	248	38.0	0.029	51	0.523	20.6%
6	index finger | middle finger | tip	247	38.0	0.092	0	0.540	0.0%
4	felt pain | pain lower | lifting	188	32.0	0.104	1	0.538	0.5%
0	haul truck | driver | loader	176	40.5	0.085	3	0.535	1.7%
2	roof fall | long wide | entry	142	39.0	0.221	0	0.583	0.0%
3	right foot | slipped fell | ankle	125	30.0	0.085	0	0.536	0.0%
1	coal rib | roof | entry	114	37.0	0.031	25	0.531	21.9%


Narratives per Cluster
7: belt | conveyor | explosion
5: fell | rock | left
8: right hand | left hand | hoist
6: index finger | middle finger | tip
4: felt pain | pain lower | lifting
0: haul truck | driver | loader
2: roof fall | long wide | entry
3: right foot | slipped fell | ankle
1: coal rib | roof | entry
0
50
100
150
200
250
300
350
400
Narrative count


Mean Document-Level Silhouette by Cluster
2: roof fall | long wide | entry
4: felt pain | pain lower | lifting
6: index finger | middle finger | tip
3: right foot | slipped fell | ankle
0: haul truck | driver | loader
1: coal rib | roof | entry
8: right hand | left hand | hoist
5: fell | rock | left
7: belt | conveyor | explosion
0.00
0.05
0.10
0.15
0.20
Mean silhouette value


Narrative Count by Cluster and Injury Severity
belt | conveyor | explosion
coal rib | roof | entry
fell | rock | left
felt pain | pain lower | lifting
haul truck | driver | loader
index finger | middle finger | tip
right foot | slipped fell | ankle
right hand | left hand | hoist
roof fall | long wide | entry
Narrative cluster
95 151 62 43 63
14 64 26 7 3
103 62 75 21 85
45 18 47 78
19 107 19 11 20
36 1 168 42
47 7 4 14 53
41 6 56 89 56
3 139
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury severity group
Narrative count
160
140
120
100
80
60
40
20
0


Injury-Severity Distribution within Each Cluster
belt | conveyor | explosion
coal rib | roof | entry
fell | rock | left
felt pain | pain lower | lifting
haul truck | driver | loader
index finger | middle finger | tip
right foot | slipped fell | ankle
right hand | left hand | hoist
roof fall | long wide | entry
Narrative cluster
23% 36% 15% 10% 15%
12% 56% 23% 6% 3%
30% 18% 22% 6% 25%
24% 10% 25% 41%
11% 61% 11% 6% 11%
15% 0% 68% 17%
38% 6% 3% 11% 42%
17% 2% 23% 36% 23%
2% 98%
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury-severity group
Percent
0
20
40
60
80

Top Terms and Phrases by Cluster

Cluster Id	Cluster Name	Term Rank	Term	Mean Tf-Idf
0	haul truck | driver | loader	1	truck	0.1245
0	haul truck | driver | loader	2	driver	0.0519
0	haul truck | driver | loader	3	loader	0.0366
0	haul truck | driver | loader	4	road	0.0306
0	haul truck | driver | loader	5	operating	0.0285
0	haul truck | driver | loader	6	highwall	0.0265
0	haul truck | driver | loader	7	investigation	0.0249
0	haul truck | driver | loader	8	haul truck	0.0243
0	haul truck | driver | loader	9	cab	0.0234
0	haul truck | driver | loader	10	fatal	0.0233
0	haul truck | driver | loader	11	load	0.0220
0	haul truck | driver | loader	12	pit	0.0215
0	haul truck | driver | loader	13	water	0.0209
0	haul truck | driver | loader	14	trailer	0.0199
0	haul truck | driver | loader	15	died	0.0198
1	coal rib | roof | entry	1	roof	0.0952
1	coal rib | roof | entry	2	entry	0.0682
1	coal rib | roof | entry	3	rib	0.0618
1	coal rib | roof | entry	4	fatal	0.0546
1	coal rib | roof | entry	5	section	0.0448
1	coal rib | roof | entry	6	crosscut	0.0445
1	coal rib | roof | entry	7	face	0.0397
1	coal rib | roof | entry	8	coal	0.0364
1	coal rib | roof | entry	9	fell	0.0355

Top Terms and Phrases by Cluster - Continued

Cluster Id	Cluster Name	Term Rank	Term	Mean Tf-Idf
1	coal rib | roof | entry	10	rock	0.0344
1	coal rib | roof | entry	11	struck	0.0333
1	coal rib | roof | entry	12	intersection	0.0323
1	coal rib | roof | entry	13	inby	0.0317
1	coal rib | roof | entry	14	coal rib	0.0301
1	coal rib | roof | entry	15	continuous	0.0300
2	roof fall | long wide | entry	1	roof fall	0.1368
2	roof fall | long wide | entry	2	fall	0.1315
2	roof fall | long wide | entry	3	entry	0.1046
2	roof fall | long wide | entry	4	long	0.0877
2	roof fall | long wide | entry	5	wide	0.0860
2	roof fall | long wide | entry	6	high	0.0712
2	roof fall | long wide | entry	7	spad	0.0622
2	roof fall | long wide | entry	8	long wide	0.0512
2	roof fall | long wide | entry	9	approx	0.0476
2	roof fall | long wide | entry	10	measured	0.0450
2	roof fall | long wide | entry	11	ft	0.0446
2	roof fall | long wide | entry	12	unintentional	0.0394
2	roof fall | long wide | entry	13	unintentional roof fall	0.0375
2	roof fall | long wide | entry	14	wide high	0.0375
2	roof fall | long wide | entry	15	bolts	0.0372
3	right foot | slipped fell | ankle	1	foot	0.1319
3	right foot | slipped fell | ankle	2	slipped	0.0945
3	right foot | slipped fell | ankle	3	right	0.0493

Top Terms and Phrases by Cluster - Continued

Cluster Id	Cluster Name	Term Rank	Term	Mean Tf-Idf
3	right foot | slipped fell | ankle	4	right foot	0.0454
3	right foot | slipped fell | ankle	5	ankle	0.0389
3	right foot | slipped fell | ankle	6	fell	0.0380
3	right foot | slipped fell | ankle	7	slipped fell	0.0363
3	right foot | slipped fell | ankle	8	knee	0.0359
3	right foot | slipped fell | ankle	9	causing	0.0352
3	right foot | slipped fell | ankle	10	foot slipped	0.0330
3	right foot | slipped fell | ankle	11	walking	0.0324
3	right foot | slipped fell | ankle	12	left	0.0266
3	right foot | slipped fell | ankle	13	boot	0.0245
3	right foot | slipped fell | ankle	14	left foot	0.0225
3	right foot | slipped fell | ankle	15	twisted	0.0207
4	felt pain | pain lower | lifting	1	felt	0.1217
4	felt pain | pain lower | lifting	2	pain	0.1175
4	felt pain | pain lower | lifting	3	felt pain	0.0588
4	felt pain | pain lower | lifting	4	lifting	0.0464
4	felt pain | pain lower | lifting	5	groin	0.0395
4	felt pain | pain lower | lifting	6	hernia	0.0342
4	felt pain | pain lower | lifting	7	shoulder	0.0342
4	felt pain | pain lower | lifting	8	lower	0.0338
4	felt pain | pain lower | lifting	9	duty	0.0291
4	felt pain | pain lower | lifting	10	pain lower	0.0284
4	felt pain | pain lower | lifting	11	sharp pain	0.0252
4	felt pain | pain lower | lifting	12	strain	0.0249

Top Terms and Phrases by Cluster - Continued

Cluster Id	Cluster Name	Term Rank	Term	Mean Tf-Idf
4	felt pain | pain lower | lifting	13	sharp	0.0247
4	felt pain | pain lower | lifting	14	diagnosed	0.0240
4	felt pain | pain lower | lifting	15	pop	0.0237
5	fell | rock | left	1	fell	0.0507
5	fell | rock | left	2	rock	0.0414
5	fell | rock | left	3	left	0.0293
5	fell | rock | left	4	ground	0.0286
5	fell | rock | left	5	knee	0.0267
5	fell | rock | left	6	struck	0.0264
5	fell | rock | left	7	piece	0.0259
5	fell | rock | left	8	head	0.0255
5	fell | rock | left	9	ankle	0.0231
5	fell | rock | left	10	right	0.0214
5	fell | rock | left	11	stepped	0.0208
5	fell | rock | left	12	hit	0.0196
5	fell | rock | left	13	walking	0.0192
5	fell | rock | left	14	striking	0.0182
5	fell | rock | left	15	leg	0.0178
6	index finger | middle finger | tip	1	finger	0.1567
6	index finger | middle finger | tip	2	index	0.0552
6	index finger | middle finger | tip	3	index finger	0.0529
6	index finger | middle finger | tip	4	tip	0.0524
6	index finger | middle finger | tip	5	middle	0.0480
6	index finger | middle finger | tip	6	middle finger	0.0439

Top Terms and Phrases by Cluster - Continued

Cluster Id	Cluster Name	Term Rank	Term	Mean Tf-Idf
6	index finger | middle finger | tip	7	left	0.0401
6	index finger | middle finger | tip	8	right	0.0373
6	index finger | middle finger | tip	9	hand	0.0367
6	index finger | middle finger | tip	10	caught	0.0323
6	index finger | middle finger | tip	11	ring	0.0312
6	index finger | middle finger | tip	12	ring finger	0.0273
6	index finger | middle finger | tip	13	left index	0.0263
6	index finger | middle finger | tip	14	right index	0.0251
6	index finger | middle finger | tip	15	amputated	0.0249
7	belt | conveyor | explosion	1	belt	0.0250
7	belt | conveyor | explosion	2	conveyor	0.0224
7	belt | conveyor | explosion	3	explosion	0.0214
7	belt | conveyor | explosion	4	eye	0.0204
7	belt | conveyor | explosion	5	causing	0.0197
7	belt | conveyor | explosion	6	material	0.0185
7	belt | conveyor | explosion	7	coal	0.0175
7	belt | conveyor | explosion	8	came	0.0150
7	belt | conveyor | explosion	9	fatally	0.0149
7	belt | conveyor | explosion	10	machine	0.0141
7	belt | conveyor | explosion	11	went	0.0135
7	belt | conveyor | explosion	12	caught	0.0135
7	belt | conveyor | explosion	13	got	0.0132
7	belt | conveyor | explosion	14	struck	0.0130
7	belt | conveyor | explosion	15	loader	0.0129

Top Terms and Phrases by Cluster - Continued

Cluster Id	Cluster Name	Term Rank	Term	Mean Tf-Idf
8	right hand | left hand | hoist	1	hand	0.0980
8	right hand | left hand | hoist	2	right hand	0.0459
8	right hand | left hand | hoist	3	hoist	0.0433
8	right hand | left hand | hoist	4	right	0.0397
8	right hand | left hand | hoist	5	elevator	0.0376
8	right hand | left hand | hoist	6	left hand	0.0342
8	right hand | left hand | hoist	7	left	0.0322
8	right hand | left hand | hoist	8	thumb	0.0309
8	right hand | left hand | hoist	9	fingers	0.0298
8	right hand | left hand | hoist	10	service	0.0227
8	right hand | left hand | hoist	11	slipped	0.0196
8	right hand | left hand | hoist	12	shaft	0.0194
8	right hand | left hand | hoist	13	door	0.0180
8	right hand | left hand | hoist	14	electrical	0.0180
8	right hand | left hand | hoist	15	hammer	0.0176

Post-Hoc Comparison with Existing MSHA Fields

Existing MSHA fields were not used to form the clusters.

The following pages show the three most common existing values within each cluster. This helps determine whether a text-derived cluster simply recreates an existing classification or connects records from several official categories.

A cluster spanning several classifications may reveal a cross-cutting event, equipment, injury, or prevention pattern.

Most Common Existing Values within Each Cluster

Cluster Id	Cluster Name	Existing Field	Rank	Existing Value	Count	Cluster Percent
0	haul truck | driver | loader	CLASSIFICATION	1	POWERED HAULAGE	89	50.6%
0	haul truck | driver | loader	CLASSIFICATION	2	SLIP OR FALL OF PERSON	18	10.2%
0	haul truck | driver | loader	CLASSIFICATION	3	MACHINERY	16	9.1%
0	haul truck | driver | loader	ACCIDENT_TYPE	1	Struck against a moving object	39	22.2%
0	haul truck | driver | loader	ACCIDENT_TYPE	2	Caught in, under or between a moving and a stationary object	32	18.2%
0	haul truck | driver | loader	ACCIDENT_TYPE	3	Fall from machine	11	6.2%
0	haul truck | driver | loader	ACTIVITY	1	Haulage or Dump truck	38	21.6%
0	haul truck | driver | loader	ACTIVITY	2	Machine maintenance	18	10.2%
0	haul truck | driver | loader	ACTIVITY	3	Utility truck, Water trucks	15	8.5%
0	haul truck | driver | loader	INJURY_SOURCE	1	HGHWY ORE CARIER,LRGE TRK	43	24.4%
0	haul truck | driver | loader	INJURY_SOURCE	2	SURFACE MINING MACHINES	24	13.6%
0	haul truck | driver | loader	INJURY_SOURCE	3	GROUND	14	8.0%
1	coal rib | roof | entry	CLASSIFICATION	1	FALL OF ROOF OR BACK	43	37.7%
1	coal rib | roof | entry	CLASSIFICATION	2	MACHINERY	20	17.5%
1	coal rib | roof | entry	CLASSIFICATION	3	POWERED HAULAGE	14	12.3%
1	coal rib | roof | entry	ACCIDENT_TYPE	1	Struck by falling object	38	33.3%
1	coal rib | roof | entry	ACCIDENT_TYPE	2	Accident type, without injuries	24	21.1%
1	coal rib | roof | entry	ACCIDENT_TYPE	3	Caught in, under or between a moving and a stationary object	21	18.4%
1	coal rib | roof | entry	ACTIVITY	1	<MISSING>	24	21.1%
1	coal rib | roof | entry	ACTIVITY	2	Continuous miner	14	12.3%
1	coal rib | roof | entry	ACTIVITY	3	Roof bolter, NEC	12	10.5%
1	coal rib | roof | entry	INJURY_SOURCE	1	CAVING ROCK,COAL,ORE,WSTE	40	35.1%

Most Common Existing Values within Each Cluster - Continued

Cluster Id	Cluster Name	Existing Field	Rank	Existing Value	Count	Cluster Percent
1	coal rib | roof | entry	INJURY_SOURCE	2	<MISSING>	25	21.9%
1	coal rib | roof | entry	INJURY_SOURCE	3	UNDERGRD MINING MACHINES	14	12.3%
2	roof fall | long wide | entry	CLASSIFICATION	1	FALL OF ROOF OR BACK	140	98.6%
2	roof fall | long wide | entry	CLASSIFICATION	2	OTHER	1	0.7%
2	roof fall | long wide | entry	CLASSIFICATION	3	FALL OF FACE/RIB/PILLAR/SIDE/HIGHWAL	1	0.7%
2	roof fall | long wide | entry	ACCIDENT_TYPE	1	Accident type, without injuries	138	97.2%
2	roof fall | long wide | entry	ACCIDENT_TYPE	2	Struck by falling object	2	1.4%
2	roof fall | long wide | entry	ACCIDENT_TYPE	3	Unclassified, insufficient data	1	0.7%
2	roof fall | long wide | entry	ACTIVITY	1	<MISSING>	138	97.2%
2	roof fall | long wide | entry	ACTIVITY	2	Unknown	2	1.4%
2	roof fall | long wide | entry	ACTIVITY	3	Timbering, Build cribs	1	0.7%
2	roof fall | long wide | entry	INJURY_SOURCE	1	<MISSING>	138	97.2%
2	roof fall | long wide | entry	INJURY_SOURCE	2	CAVING ROCK,COAL,ORE,WSTE	3	2.1%
2	roof fall | long wide | entry	INJURY_SOURCE	3	MISCELLANEOUS,NEC	1	0.7%
3	right foot | slipped fell | ankle	CLASSIFICATION	1	SLIP OR FALL OF PERSON	72	57.6%
3	right foot | slipped fell | ankle	CLASSIFICATION	2	HANDLING OF MATERIALS	16	12.8%
3	right foot | slipped fell | ankle	CLASSIFICATION	3	POWERED HAULAGE	10	8.0%
3	right foot | slipped fell | ankle	ACCIDENT_TYPE	1	Over-exertion NEC	25	20.0%
3	right foot | slipped fell | ankle	ACCIDENT_TYPE	2	Fall to the walkway or working surface	21	16.8%
3	right foot | slipped fell | ankle	ACCIDENT_TYPE	3	Struck by falling object	15	12.0%
3	right foot | slipped fell | ankle	ACTIVITY	1	Walking, Running	31	24.8%
3	right foot | slipped fell | ankle	ACTIVITY	2	Handling supplies or material	22	17.6%

Most Common Existing Values within Each Cluster - Continued

Cluster Id	Cluster Name	Existing Field	Rank	Existing Value	Count	Cluster Percent
3	right foot | slipped fell | ankle	ACTIVITY	3	Machine maintenance	20	16.0%
3	right foot | slipped fell | ankle	INJURY_SOURCE	1	GROUND	26	20.8%
3	right foot | slipped fell | ankle	INJURY_SOURCE	2	FLOOR,WALKING SURF-NOT UG	12	9.6%
3	right foot | slipped fell | ankle	INJURY_SOURCE	3	METAL COVERS & GUARDS	10	8.0%
4	felt pain | pain lower | lifting	CLASSIFICATION	1	HANDLING OF MATERIALS	117	62.2%
4	felt pain | pain lower | lifting	CLASSIFICATION	2	OTHER	15	8.0%
4	felt pain | pain lower | lifting	CLASSIFICATION	3	SLIP OR FALL OF PERSON	14	7.4%
4	felt pain | pain lower | lifting	ACCIDENT_TYPE	1	Over-exertion in lifting objects	62	33.0%
4	felt pain | pain lower | lifting	ACCIDENT_TYPE	2	Over-exertion NEC	62	33.0%
4	felt pain | pain lower | lifting	ACCIDENT_TYPE	3	Over-exertion in pulling or pushing objects	19	10.1%
4	felt pain | pain lower | lifting	ACTIVITY	1	Handling supplies or material	69	36.7%
4	felt pain | pain lower | lifting	ACTIVITY	2	Machine maintenance	19	10.1%
4	felt pain | pain lower | lifting	ACTIVITY	3	Hand load, Hand shoveling	12	6.4%
4	felt pain | pain lower | lifting	INJURY_SOURCE	1	MISCELLANEOUS,NEC	22	11.7%
4	felt pain | pain lower | lifting	INJURY_SOURCE	2	BROKEN ROCK,COAL,ORE,WSTE	18	9.6%
4	felt pain | pain lower | lifting	INJURY_SOURCE	3	METAL,NEC(PIPE,WIRE,NAIL)	16	8.5%
5	fell | rock | left	CLASSIFICATION	1	SLIP OR FALL OF PERSON	84	24.3%
5	fell | rock | left	CLASSIFICATION	2	HANDLING OF MATERIALS	52	15.0%
5	fell | rock | left	CLASSIFICATION	3	DISORDERS (REPEATED TRAUM.	44	12.7%
5	fell | rock | left	ACCIDENT_TYPE	1	Struck by falling object	67	19.4%
5	fell | rock | left	ACCIDENT_TYPE	2	Over-exertion NEC	45	13.0%
5	fell | rock | left	ACCIDENT_TYPE	3	NEC	29	8.4%

Most Common Existing Values within Each Cluster - Continued

Cluster Id	Cluster Name	Existing Field	Rank	Existing Value	Count	Cluster Percent
5	fell | rock | left	ACTIVITY	1	Machine maintenance	48	13.9%
5	fell | rock | left	ACTIVITY	2	Walking, Running	48	13.9%
5	fell | rock | left	ACTIVITY	3	Handling supplies or material	44	12.7%
5	fell | rock | left	INJURY_SOURCE	1	CAVING ROCK,COAL,ORE,WSTE	38	11.0%
5	fell | rock | left	INJURY_SOURCE	2	GROUND	35	10.1%
5	fell | rock | left	INJURY_SOURCE	3	NOISE,NEC	26	7.5%
6	index finger | middle finger | tip	CLASSIFICATION	1	HANDLING OF MATERIALS	121	49.0%
6	index finger | middle finger | tip	CLASSIFICATION	2	MACHINERY	59	23.9%
6	index finger | middle finger | tip	CLASSIFICATION	3	HANDTOOLS (NONPOWERED)	33	13.4%
6	index finger | middle finger | tip	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	104	42.1%
6	index finger | middle finger | tip	ACCIDENT_TYPE	2	Caught in, under or between NEC	60	24.3%
6	index finger | middle finger | tip	ACCIDENT_TYPE	3	Struck by... NEC	28	11.3%
6	index finger | middle finger | tip	ACTIVITY	1	Machine maintenance	78	31.6%
6	index finger | middle finger | tip	ACTIVITY	2	Handling supplies or material	63	25.5%
6	index finger | middle finger | tip	ACTIVITY	3	Hand tools (not powered)	33	13.4%
6	index finger | middle finger | tip	INJURY_SOURCE	1	METAL,NEC(PIPE,WIRE,NAIL)	54	21.9%
6	index finger | middle finger | tip	INJURY_SOURCE	2	METAL COVERS & GUARDS	35	14.2%
6	index finger | middle finger | tip	INJURY_SOURCE	3	BROKEN ROCK,COAL,ORE,WSTE	13	5.3%
7	belt | conveyor | explosion	CLASSIFICATION	1	HANDLING OF MATERIALS	89	21.5%
7	belt | conveyor | explosion	CLASSIFICATION	2	MACHINERY	75	18.1%
7	belt | conveyor | explosion	CLASSIFICATION	3	POWERED HAULAGE	54	13.0%
7	belt | conveyor | explosion	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	52	12.6%

Most Common Existing Values within Each Cluster - Continued

Cluster Id	Cluster Name	Existing Field	Rank	Existing Value	Count	Cluster Percent
7	belt | conveyor | explosion	ACCIDENT_TYPE	2	Accident type, without injuries	44	10.6%
7	belt | conveyor | explosion	ACCIDENT_TYPE	3	Struck by... NEC	30	7.2%
7	belt | conveyor | explosion	ACTIVITY	1	Machine maintenance	75	18.1%
7	belt | conveyor | explosion	ACTIVITY	2	Handling supplies or material	63	15.2%
7	belt | conveyor | explosion	ACTIVITY	3	<MISSING>	44	10.6%
7	belt | conveyor | explosion	INJURY_SOURCE	1	<MISSING>	44	10.6%
7	belt | conveyor | explosion	INJURY_SOURCE	2	METAL,NEC(PIPE,WIRE,NAIL)	35	8.5%
7	belt | conveyor | explosion	INJURY_SOURCE	3	MISCELLANEOUS,NEC	25	6.0%
8	right hand | left hand | hoist	CLASSIFICATION	1	HANDLING OF MATERIALS	76	30.6%
8	right hand | left hand | hoist	CLASSIFICATION	2	HOISTING	52	21.0%
8	right hand | left hand | hoist	CLASSIFICATION	3	MACHINERY	41	16.5%
8	right hand | left hand | hoist	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	66	26.6%
8	right hand | left hand | hoist	ACCIDENT_TYPE	2	Accident type, without injuries	51	20.6%
8	right hand | left hand | hoist	ACCIDENT_TYPE	3	Struck by... NEC	34	13.7%
8	right hand | left hand | hoist	ACTIVITY	1	Machine maintenance	65	26.2%
8	right hand | left hand | hoist	ACTIVITY	2	<MISSING>	51	20.6%
8	right hand | left hand | hoist	ACTIVITY	3	Handling supplies or material	34	13.7%
8	right hand | left hand | hoist	INJURY_SOURCE	1	<MISSING>	51	20.6%
8	right hand | left hand | hoist	INJURY_SOURCE	2	METAL,NEC(PIPE,WIRE,NAIL)	29	11.7%
8	right hand | left hand | hoist	INJURY_SOURCE	3	METAL COVERS & GUARDS	27	10.9%

Cluster 0: haul truck | driver | loader - 1 of 2

Cluster ID: 0
Provisional label: haul truck | driver | loader
Narratives: 176
Mean document silhouette: 0.085
Negative-silhouette narratives: 3 (1.7%)

TOP TERMS AND PHRASES
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truck, driver, loader, road, operating, highwall, investigation, haul truck, cab, fatal, load, pit, water, trailer, died

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220232580014 | Fatality | Silhouette 0.220 | Distance 0.671
Unknown- Truck over traveled dump site through berm. Truck traveled down and resting submerged in pit. Accident under investigation.

Representative 2 | Record 220141110020 | No lost time or other | Silhouette 0.187 | Distance 0.694
A blaster for sub-contractor, Austin Powder, left his truck in reverse gear and let truck roll off 14-16'high embankment. No injuries, removed truck and drove out of mine site, attached to wrecker, and went to Austin Powder shop.

Representative 3 | Record 220051610057 | Fatality | Silhouette 0.205 | Distance 0.711
TRAINING IN A TRUCK WHERE THE DRIVER STATES THAT HE LOST CONTROL OF THE EMPTY CAT 773 DUMP TRUCK. THE TRUCK VEERED, RAN THROUGH THE PROTECTIVE BERM AND OFF A 70'-HIGH WALL. (TRUCK LANDED ON ITS TOP; BLUNT FORCE TRAUMA INSIDE CAB OF TRUCK.)

Representative 4 | Record 220132660033 | Fatality | Silhouette 0.168 | Distance 0.713
temperature were around 70 degrees, sunny and dry day. Adding to narrative per prelim **The victim was operating a 2 1/2 ton truck up a steep roadway. He was hauling water tanks on the bed of the truck when the truck overturned on its roof and crushed him.**

Representative 5 | Record 220240380015 | Fatality | Silhouette 0.206 | Distance 0.715
Truck driver was found unresponsive in the cab of EE's overturned truck at the bottom of the feed ramp drop off. There were no witnesses to the accident. Investigation is still pending.

BORDERLINE NARRATIVES

Borderline 1 | Record 220071160024 | Fatality | Silhouette -0.048 | Distance 0.970
A highwall measuring approx 150' long, 180' high, & 75' wide collapsed resulting in fatal injuries to 2 miners that were working beneath the highwall. One was operating an excavator & the other was operating a bulldozer. Examination of the remaining highwall indicates the highwall failure originated at the toe of the columnar section above the Pittsburgh Big Vein coal seam.

Borderline 2 | Record 220151610008 | Fatality | Silhouette -0.016 | Distance 0.903
Employee was operating a diesel personnel carrier on the Main South Travelway. It appears that his vehicle struck a coal pillar at xcut #193 and he was ejected from the vehicle. He sustained fatal crushing injuries to his chest area.

INTERPRETATION NOTE

Cluster 0: haul truck | driver | loader - 2 of 2

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed group.

Cluster 1: coal rib | roof | entry - 1 of 2

Cluster ID: 1
Provisional label: coal rib | roof | entry
Narratives: 114
Mean document silhouette: 0.031
Negative-silhouette narratives: 25 (21.9%)

TOP TERMS AND PHRASES
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roof, entry, rib, fatal, section, crosscut, face, coal, fell, rock, struck, intersection, inby, coal rib, continuous

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220033070001 | Fatality | Silhouette 0.015 | Distance 0.718
A FATAL ROOF FALL OCCURRED TO EE AS HE WAS RUNNING THRU AN INTERSECTION OF #3 ENTRY IN THE 1 LEFT SECTION. HE HAD BEEN SETTING BREAKER POSTS ON THE PILLARLINE WHEN HE HEARD THE TOP RUMBLING AN D HAD MOVED OUTBY ONE BLOCK AND STARTED THRU THECROSSCUT WHEN REACHING THE INTERSECTION, THE ROOF COLLAPSED ON HIM.

Representative 2 | Record 220053630016 | Fatality | Silhouette 0.081 | Distance 0.728
While in the process of installing permanent roof support in the No. 1 Entry, the roof started to fall and the employee ran inby the ATRS System. The roof fell in this area causing crushing injuries to the employee.

Representative 3 | Record 220051730008 | Fatality | Silhouette 0.102 | Distance 0.747
The victim had traveled inby permanent roof support in the #3 entry right crosscut #4 and a section of draw rock fell in the unsupported area fatally injuring the victim. The continuous miner was cutting into #3 right crosscut from the #4 entry when the draw rock fell out on miner and victim.

Representative 4 | Record 220091830023 | Fatality | Silhouette 0.092 | Distance 0.758
Employee was in the process of setting or had set a roof jack, on the left inby corner or #60 Crosscut between the No 2 and No. 1 entry in the 4 west miner section. A piece of stone fell out of the roof and struck the employee, pinning him to the ground.

Representative 5 | Record 220045550003 | Fatality | Silhouette 0.051 | Distance 0.765
A FATAL FALL OF ROOF ACCIDENT OCCURRED ON THE 003 WORKING SECTION. THE VICTIM WAS WORKING WITH THE SECTION CREW AND HAD BEEN INSTALLING TIMBERS DURING RETREAT MINING. THE CREW HAD TEMPORARILY HALTED PRODUCTION TO OBSERVE THE MINE ROOF, WHEN WITH LITTLE WARNING THE MINE ROOF BEGAN TO COLLAPSE. THE 5 MINERS RAN TO ESCAPE THE ROOF FALL. FOUR OF THE MINERS ESCAPED.

BORDERLINE NARRATIVES

Borderline 1 | Record 220122680014 | No lost time or other | Silhouette -0.102 | Distance 0.847
The mains section had advanced under a hollow. The #9 intersection began to take weight. The # 9 return entry at 1 break inby spad # 2112 was cribbed off. no effort was made to save the intersection. The intersection in the #8 return entry was reinforced with cribs, steel beams, jacks and longer bolts.

Borderline 2 | Record 220151590007 | No lost time or other | Silhouette -0.096 | Distance 0.881
A roof fall occurred in 1 west bleeder at 84xc. Was approx. 8' H, 18' W and 25' L. Area was bolted with 8' point anchor resin assisted bolts, 12' CC cable bolts in pattern along with roof screen and bacon

Cluster 1: coal rib | roof | entry - 2 of 2

straps. Fall did not obstruct vent. or escapeways. Cause of fall was deterioration due to weathering. Floor/Roof support will be set as reported to prevent future occurrences.

INTERPRETATION NOTE

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed group.

Cluster 2: roof fall | long wide | entry - 1 of 2

Cluster ID: 2
Provisional label: roof fall | long wide | entry
Narratives: 142
Mean document silhouette: 0.221
Negative-silhouette narratives: 0 (0.0%)

TOP TERMS AND PHRASES
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roof fall, fall, entry, long, wide, high, spad, long wide, approx, measured, ft, unintentional, unintentional roof fall, wide high, bolts

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220000180085 | No lost time or other | Silhouette 0.321 | Distance 0.464
A ROOF FALL OCCURRED IN THE #1 ENTRY OF THE #2 LEFT PANEL MEASURING 18' WIDE X 210' LONG X 8' HIGH. THE FALL BEGAN 10 FT INBY SPAD #1866 EXTENDING 3 CROSSCUTS INBY AND THROUGH THE CROSSCUT TO THE #2 ENTRY. NO EQUIPMENT WAS INVOLVED. THE VENTILATION WAS NOT DISTURBED. THE WAS NO INJURY INVOLVED. THE FALL WILL NOT BE CLEANED UP.

Representative 2 | Record 220033280062 | No lost time or other | Silhouette 0.356 | Distance 0.470
A ROOF FALL WAS DISCOVERED IN THE OLD #2 UNIT PANEL, #6 ENTRY, INBY STATION SPAD #5915 APPROXIMATELY 30'. THE FALL MEASURES 30' LONG, 20' WIDE, & 7' HIGH.

Representative 3 | Record 220222850004 | No lost time or other | Silhouette 0.327 | Distance 0.482
An unintentional roof fall occurred in the return of MMU 001 on 3 north 8 panel. It fell from 1 thru 5 entry 30 feet outby S/S 10720 in #3 entry. The roof fall measured approx. 18' wide by 250 long at a undetermined height. The exact cause of the fall is unknown but due to other caves occurring in this area previously the operator is abandoning any further mining in this area.

Representative 4 | Record 220022620003 | No lost time or other | Silhouette 0.329 | Distance 0.489
ROOF FALL WAS DISCOVERED ON 9-14-02 AT THE #3 ENTRY OF SS# 10258 APPROX. 6000'OUTBY THE WORKING SECTION. THE FALL MEASURED 24'LONG X 18'WIDE X 6'THICK IN THE INTERSECTION. NO EQUIPMENT INVOLV ED, NO VENTILATION INTERRUPTED, FALL WILL NOT BECLEANED UP.

Representative 5 | Record 220012850080 | No lost time or other | Silhouette 0.323 | Distance 0.492
AN UNINTENTIONAL ROOF FALL OCCURRED IN THE INTERSECTION OF #3 ENTRY, 8 BLOCKS OUT BY THE FACE. THE AREA HAD SOME SLICK SIDE CONDITIONS IN IT AS IT WAS MINED. FALL MEASURED 20' LONG BY 17' WID E BY 10' HIGH, BOLTED WITH 5' RESIN 6 BY 16 PLATES.

BORDERLINE NARRATIVES

Borderline 1 | Record 220090620023 | No lost time or other | Silhouette 0.098 | Distance 0.857
A roof fall was discovered at 4 mains 21x-cut #14hdg. This fall was 1,600 feet out-by the feeder. Fell above the anchorage of six foot and eight foot super twist roof bolts. The area was supplemented with 12 foot post tensioned cable bolts. Fall was in the intersection only. approx. 18' by 18' in area.

Borderline 2 | Record 220031680030 | No lost time or other | Silhouette 0.099 | Distance 0.860
FOUND A ROOF FALL #9 ENTRY INTERSECTION OUTBY FACE.

INTERPRETATION NOTE

Cluster 2: roof fall | long wide | entry - 2 of 2

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed group.

Cluster 3: right foot | slipped fell | ankle

Cluster ID: 3
Provisional label: right foot | slipped fell | ankle
Narratives: 125
Mean document silhouette: 0.085
Negative-silhouette narratives: 0 (0.0%)

TOP TERMS AND PHRASES
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foot, slipped, right, right foot, ankle, fell, slipped fell, knee, causing, foot slipped, walking, left, boot, left foot, twisted

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220231030025 | Days away | Silhouette 0.154 | Distance 0.703
While exiting 1394 Haul Truck, Left hand holding rail, right foot stepped down on next step but foot slipped. Landed on buttocks on the bottom of last step.

Representative 2 | Record 220001880009 | No lost time or other | Silhouette 0.190 | Distance 0.705
COMPANY TEAM LEADERS WAS HAVING A BONDING SHARING EXPERIENCE AT ZION'S PARK. WE WERE HIKING UP ZION'S NARROWS WHEN INJURED'S FOOT SLIPPED ON A SLICK ROCK AND BROKE HER FOOT.

Representative 3 | Record 220112020028 | Restricted or transferred | Silhouette 0.183 | Distance 0.707
Employee was running out rods and slipped and fell dropping rod on foot.

Representative 4 | Record 220040410012 | Restricted or transferred | Silhouette 0.162 | Distance 0.719
EMPLOYEE EXITED PICKUP, CLOSED DOOR, AND TURNED TO GO TO THE OFFICE. HIS RIGHT FOOT SLIPPED & CAUGHT, CAUSING ANKLE PAIN.

Representative 5 | Record 220052620052 | Days away | Silhouette 0.170 | Distance 0.738
Employee stated he was going through a mandoor and his foot slipped and he injured his back. Employee has a history of previous back problems and we are currently investigating this claim. Employee first lost work day did not occur until 9/6/05.

BORDERLINE NARRATIVES

Borderline 1 | Record 220002490102 | Restricted or transferred | Silhouette 0.001 | Distance 0.943
EMPLOYEE WAS CARRYING A 2' LONG 8X8 POST TO THE LOCATION WHERE HE WAS WORKING; THE POST WAS WET DUE TO RECENT RAINFALL IN THE AREA AND SLIPPED FROM HIS HANDS. HE BROKE HIS FINGER WHEN HE TRIE D TO CATCH THE POST BEFORE IT HIT THE GROUND.

Borderline 2 | Record 220013170126 | Permanent disability | Silhouette 0.005 | Distance 0.959
CHANGING PULLEY BELT FINGER ON PULLEY SLIPPED & FELL.

INTERPRETATION NOTE

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed group.

Cluster 4: felt pain | pain lower | lifting

Cluster ID: 4
Provisional label: felt pain | pain lower | lifting
Narratives: 188
Mean document silhouette: 0.104
Negative-silhouette narratives: 1 (0.5%)

TOP TERMS AND PHRASES
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felt, pain, felt pain, lifting, groin, hernia, shoulder, lower, duty, pain lower, sharp pain, strain, sharp, diagnosed, pop

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220002560029 | Permanent disability | Silhouette 0.217 | Distance 0.709
EMPLOYEE WAS LIFTING A CONCRETE FORM (8' BY 2') BY HIMSELF WHEN HE FELT A PAIN IN HIS STOMACH.

Representative 2 | Record 220160190033 | Restricted or transferred | Silhouette 0.186 | Distance 0.711
Initial injury occurred on 6 Jan 2016 and not reported to operator supervisor until 11 Jan 2016 after employee felt they needed to see physician for groin pain. Individual was removing sand from work area with a large shovel on main floor of 320 silo when initial discomfort was noticed. After the 5 days, the individual felt they needed to visit doctor.

Representative 3 | Record 220002490149 | Permanent disability | Silhouette 0.209 | Distance 0.713
WHILE PUTTING OIL THE FORKLIFT THE EE STRETCHED AND FELT PAIN IN THE STOMACH AREA. (EE CONTINUED WORKING UNTIL 8-21-00--STARTED MISSING WORK ON 8-21-00)

Representative 4 | Record 220010360156 | Permanent disability | Silhouette 0.202 | Distance 0.718
WHILE TRANSFERRING DIRT FROM BUCKET TO BOX HE FELT PAIN IN GROIN AREA. STARTED MISSING WORK ON 1/16/01.

Representative 5 | Record 220001470041 | Permanent disability | Silhouette 0.202 | Distance 0.720
INJURED FELT PAIN IN HIS RIGHT GROIN WHILE LIFTING A RETURN ROLLER. (DR. TOOK HIM OFF WORK WITH A HERNIA ON 5-15-2000).

BORDERLINE NARRATIVES

Borderline 1 | Record 220000210018 | Days away | Silhouette -0.003 | Distance 0.956
EE WAS LOOSENING BOLT ON ENGINE REMOVAL ON D9 DOZER.. TOOL SLIPPED OFF NUT, EE LOST HIS BALANCE, FALLING TO GROUND. PAIN ON RIGHT SIDE--RIB AND SHOULDER BRUISED.

Borderline 2 | Record 220211870021 | Restricted or transferred | Silhouette 0.014 | Distance 0.909
A maintenance craftsman was climbing onto the boarding ladder of an electric shovel when they felt a pop in their right foot. The employee was seen at a medical facility and given restricted duty pending a follow up appointment.

INTERPRETATION NOTE

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed group.

Cluster 5: fell | rock | left - 1 of 2

Cluster ID: 5
Provisional label: fell | rock | left
Narratives: 346
Mean document silhouette: -0.005
Negative-silhouette narratives: 190 (54.9%)

TOP TERMS AND PHRASES
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fell, rock, left, ground, knee, struck, piece, head, ankle, right, stepped, hit, walking, striking, leg

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220043500002 | Days away | Silhouette 0.025 | Distance 0.860
Employee was "dredging" water away from a piece of equipment with his right heel when his left leg gave away and he fell to the ground. Two bones were broken above the ankle.

Representative 2 | Record 220101900010 | Days away | Silhouette 0.006 | Distance 0.863
The IE, a coal hauler operator, was found laying on the ground holding himself up by his left elbow. The IE was in xc 2-1 at station # 21+15, when he was struck by a rock, that fell out between the roof bolts, and struck him on top of the head. The rock measured 10'L x 3'W x 0-11" thick, and fell from a height of 7'11".

Representative 3 | Record 220262440018 | Days away | Silhouette 0.027 | Distance 0.864
A mechanic was using crosswalk and fell to the ground landing on their left knee and elbow. They employee did not go on restrictions or lost time but did start physical therapy 08/24/2026.

Representative 4 | Record 220060970021 | Fatality | Silhouette -0.005 | Distance 0.870
While installing roof pins in front of LW shield, piece of rock fell, striking employee and knocking him down. Contusion to back of head, laceration to right hand and fracture to left ankle. The victim was flown by helicopter to the nearest trauma center, where he died on April 10, 2006.

Representative 5 | Record 220261610005 | Days away | Silhouette 0.041 | Distance 0.875
Employee alleges EE dislodged rock from conveyer and when rock gave way, EE fell to ground and strained ankle

BORDERLINE NARRATIVES

Borderline 1 | Record 220071980032 | Days away | Silhouette -0.106 | Distance 0.933
Employee fell thru a elevated walkway when the steel grating approx. 22" wide and 36" long he stepped on dislodged. the walkway was supported on both side with 3"x3" angle iron and was adjacent to a crusher. It appears the weld broke as a result of vibration.

Borderline 2 | Record 220013620011 | Permanent disability | Silhouette -0.099 | Distance 0.936
EMPLOYEE WAS OPERATING MINER, CLEANING RIGHT HAND CROSSCUT IN #3 ENTRY WHEN A ROCK FELL STRIKING HIM. THE ROCK MEASURED 36"LONG, 18"WIDE, 6"THICK.

INTERPRETATION NOTE

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed

Cluster 5: fell | rock | left - 2 of 2

group.

Cluster 6: index finger | middle finger | tip

Cluster ID: 6
Provisional label: index finger | middle finger | tip
Narratives: 247
Mean document silhouette: 0.092
Negative-silhouette narratives: 0 (0.0%)

TOP TERMS AND PHRASES
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finger, index, index finger, tip, middle, middle finger, left, right, hand, caught, ring, ring finger, left index, right index, amputated

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220230890016 | Permanent disability | Silhouette 0.194 | Distance 0.680
A miner and partner were moving a 10"X10" cap. As they were setting the cap in place one of the miners caught the tip of left index finger. The tip of the finger was de-gloved. Surgery was preformed the following day. Amputation to the first knuckle was required.

Representative 2 | Record 220002310036 | Permanent disability | Silhouette 0.160 | Distance 0.694
ON SOUTH END HALF DECK OF PRESS FLOOR, TAKING DOWN AN AIR BLOW DOWN POT, WHEN IT ROLLED AND JAMMED HIS RIGHT INDEX FINGER AGAINST THE MID BAR OF THE HAND RAIL. DISTAL TIP OF FINGER AMPUTATED.

Representative 3 | Record 220112910049 | Permanent disability | Silhouette 0.182 | Distance 0.701
He was helping lay track and caught his right index finger and middle finger between the bottom of a track tie and the bottom and cut the tips off each finger.

Representative 4 | Record 220171160040 | Permanent disability | Silhouette 0.164 | Distance 0.715
Employee was putting a bearing on a mixer with hand. The bearing moved quicker then expected and ee smashed finger between the bearing and the mixer. The tip of left index finger was amputated by dr.

Representative 5 | Record 220213410008 | Permanent disability | Silhouette 0.143 | Distance 0.715
Employee was cleaning inside of a Reed Gunite Pot and got left hand glove caught between feed wheel and rock shear. Severed tip of middle finger and index finger.

BORDERLINE NARRATIVES

Borderline 1 | Record 220013510004 | Restricted or transferred | Silhouette 0.021 | Distance 0.977
EE WAS BOLTING ON MINER. HE WAS ATTEMPTING TO INSERT DRILL STEEL INTO CHUCK AS HE RAISED THE DRILL HEAD. THE STEEL CAUGHT AGAINST THE ROOF & BENT, CATCHING HIS L. LITTLE FINGER BETWEEN THE DRILL STEEL AND THE BOLTER MAST.

Borderline 2 | Record 220051400016 | Days away | Silhouette 0.024 | Distance 0.942
Member, was moving rt. roof bolter when atrs extension came out its slots. When moving it back in place the end came up smashing finger on roof.

INTERPRETATION NOTE

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed group.

Cluster 7: belt | conveyor | explosion - 1 of 2

Cluster ID: 7
Provisional label: belt | conveyor | explosion
Narratives: 414
Mean document silhouette: -0.009
Negative-silhouette narratives: 261 (63.0%)

TOP TERMS AND PHRASES
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belt, conveyor, explosion, eye, causing, material, coal, came, fatally, machine, went, caught, got, struck, loader

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220060130058 | No lost time or other | Silhouette 0.033 | Distance 0.889
A loose connection on a transformer caused a power failure to the plant air compressor which caused exhaust valves to close on the Pyro main baghouse which allowed hot gases to reach the dryer feed belts and started a fire. The Mobile Fire Dept was called and they put the fire out in less than an hour. Valves will be installed to prevent future like incidents

Representative 2 | Record 220242180002 | Fatality | Silhouette 0.037 | Distance 0.897
While cleaning raw coal material from on top of the locked out Raw Coal Stacking Conveyor, the Miner was standing on the conveyor ahead of the conveyor feed chute. The belt and material moved backwards, sweeping the Miner into the chute and covered Miner with material.

Representative 3 | Record 220051710073 | Days away | Silhouette 0.031 | Distance 0.899
EE WAS WELDING ON A CONVEYOR AND AFTER A TIME HE USED A FORKLIFT TO MOVE THE CONVEYOR. WHEN HE RETURNED TO CONTINUE WELDING HE FORGOT HE HAD MOVED THE CONVEYOR. WHEN HE TURNED AROUND TO PUT ON HIS WELDING SHIELD HELMET, HE BUMPED HIS FOREHEAD AGAINST FORKTRUCK FORK, CAUSING HIM TO CLENCH HIS TEETH. THIS CAUSED A TOOTH TO BE DAMAGED.

Representative 4 | Record 220131700022 | Days away | Silhouette 0.036 | Distance 0.905
Employee needed to repair a broken wire on a D-11 dozer that was located under cab floor. As employee was in this area he pushed against a hydraulic line, this caused dirt to fall from the hydraulic line onto his forehead and down on the inside of his safety glasses. Some foreign material got into the employee's right eye.

Representative 5 | Record 220242320035 | Fatality | Silhouette 0.019 | Distance 0.906
Employee was shoveling coal off a conveyor belt, belt moved forward causing miner to fall down transfer chute onto belt conveyor that transferred EE to stacker tube where EE fell onto coal pile.

BORDERLINE NARRATIVES

Borderline 1 | Record 220092430086 | Fatality | Silhouette -0.106 | Distance 0.983
The individual was struck by a rib roll measuring approximately 20' long X 2' thick X 4 1/2' wide causing crushing injuries.

Borderline 2 | Record 220112490009 | No lost time or other | Silhouette -0.101 | Distance 0.969
Grass fire started in non active area of mine, due to tree contacting high line wires during high winds. Approximately two acres affected.

Cluster 7: belt | conveyor | explosion - 2 of 2

INTERPRETATION NOTE

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed group.

Cluster 8: right hand | left hand | hoist - 1 of 2

Cluster ID: 8
Provisional label: right hand | left hand | hoist
Narratives: 248
Mean document silhouette: 0.029
Negative-silhouette narratives: 51 (20.6%)

TOP TERMS AND PHRASES
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hand, right hand, hoist, right, elevator, left hand, left, thumb, fingers, service, slipped, shaft, door, electrical, hammer

REPRESENTATIVE NARRATIVES

Representative 1 | Record 220040790050 | Permanent disability | Silhouette 0.072 | Distance 0.795
crushed hand resulting in left thumb amputation at Ip joint. Distal fractures fingers 2,3,4. Large laceration to palm of left hand.

Representative 2 | Record 220221540013 | Permanent disability | Silhouette 0.013 | Distance 0.800
While overseeing a maintenance job, the manager was checking the crews progress of reinstalling the impeller. The manager placed a hand on the pump shell to look around the back side. The hoist tool shifted onto manager's hand causing crushing injury to left ring finger.

Representative 3 | Record 220071140049 | Permanent disability | Silhouette 0.069 | Distance 0.803
Employee was placing the dishcarge trunion bearing into place on Mill #6. He was trying to remove a piece of cribbing when the cribbing shifted pinching fingers on his right hand between the cribbing (wood) and the casing on the mill. Resulted in laceration and partial amputation of mid and ring fingers of right hand.

Representative 4 | Record 220200580012 | Permanent disability | Silhouette 0.013 | Distance 0.804
Employee received partial fingertip amputation, crush injury to finger and open fracture of distal phalanx of right hand when positioning forks on a Cat 1255D Telehandler. Forks were pushed out by employee who had a hand behind forks and when they swung back, fingers were caught between forks and mast.

Representative 5 | Record 220253170158 | Restricted or transferred | Silhouette 0.046 | Distance 0.807
Injured MTM was sitting in a spare 3rd stage mainframe in the zinc shop installing a horse collar liner. EE was operating the overhead crane with EE's right hand and repositioning a horse collar liner for final placement with EE's left hand when EE's left fingertips became pinched between the bottom of the tub and the liner. This resulted in a fractured left middle fingertip

BORDERLINE NARRATIVES

Borderline 1 | Record 220130030030 | Restricted or transferred | Silhouette -0.050 | Distance 0.881
Installing hub seal with another employee. Repositioned hand when dead blow hammer stuck left ring finger resulting in fracture and stitches.

Borderline 2 | Record 220021090046 | Restricted or transferred | Silhouette -0.046 | Distance 0.862
PUTTING PALLETS ON PALLET LIFT, SMASHED RING FINGER ON LEFT HAND.

INTERPRETATION NOTE

Cluster 8: right hand | left hand | hoist - 2 of 2

The provisional label is generated from cluster vocabulary. The representative narratives should be used to decide whether the cluster reflects one coherent event pattern, several related patterns, or a mixed group.

Candidate Corpus Vocabulary

The candidate vocabulary excludes standard English stop words and common narrative boilerplate.

Terms must occur in at least three documents, but not in more than 70% of the sampled narratives.

The vocabulary score gives a modest preference to two-word phrases because phrases often retain more operational context than isolated words.

These terms are candidates for interpreting clusters. They are not automatically accepted as the final project vocabulary.

Top Candidate Vocabulary Terms and Phrases

Term Or Phrase	Documents	Document Prevalence	Mean Tf-Idf	Phrase Length	Vocabulary Score
finger	298	14.9%	0.0223	1	0.0223
left	360	18.0%	0.0220	1	0.0220
right	347	17.3%	0.0220	1	0.0220
hand	300	15.0%	0.0205	1	0.0205
fell	256	12.8%	0.0186	1	0.0186
fall	182	9.1%	0.0159	1	0.0159
rock	152	7.6%	0.0156	1	0.0156
causing	219	10.9%	0.0149	1	0.0149
roof fall	118	5.9%	0.0107	2	0.0139
slipped	176	8.8%	0.0138	1	0.0138
belt	138	6.9%	0.0130	1	0.0130
truck	123	6.2%	0.0129	1	0.0129
felt	143	7.1%	0.0127	1	0.0127
entry	144	7.2%	0.0124	1	0.0124
pain	133	6.7%	0.0123	1	0.0123
struck	148	7.4%	0.0123	1	0.0123
caught	141	7.0%	0.0108	1	0.0108
foot	105	5.2%	0.0106	1	0.0106
right hand	92	4.6%	0.0077	2	0.0100
piece	111	5.5%	0.0098	1	0.0098
using	119	5.9%	0.0094	1	0.0094
ground	100	5.0%	0.0093	1	0.0093
left hand	83	4.2%	0.0071	2	0.0092
went	110	5.5%	0.0092	1	0.0092
head	97	4.9%	0.0092	1	0.0092

Top Candidate Vocabulary Terms and Phrases - Continued

Term Or Phrase	Documents	Document Prevalence	Mean Tf-Idf	Phrase Length	Vocabulary Score
knee	80	4.0%	0.0091	1	0.0091
got	102	5.1%	0.0090	1	0.0090
loader	71	3.5%	0.0088	1	0.0088
cut	99	5.0%	0.0087	1	0.0087
index finger	78	3.9%	0.0067	2	0.0087
tip	106	5.3%	0.0084	1	0.0084
roof	83	4.2%	0.0084	1	0.0084
lower	85	4.2%	0.0082	1	0.0082
section	79	4.0%	0.0081	1	0.0081
long	99	5.0%	0.0080	1	0.0080
did	90	4.5%	0.0080	1	0.0080
operating	82	4.1%	0.0079	1	0.0079
middle	86	4.3%	0.0078	1	0.0078
approx	85	4.2%	0.0078	1	0.0078
pipe	56	2.8%	0.0076	1	0.0076
hit	75	3.8%	0.0075	1	0.0075
index	89	4.5%	0.0075	1	0.0075
middle finger	60	3.0%	0.0057	2	0.0074
came	83	4.2%	0.0074	1	0.0074
conveyor	60	3.0%	0.0074	1	0.0074
coal	68	3.4%	0.0074	1	0.0074
material	70	3.5%	0.0074	1	0.0074
face	72	3.6%	0.0073	1	0.0073
felt pain	46	2.3%	0.0056	2	0.0073
hammer	57	2.9%	0.0073	1	0.0073

Top Candidate Vocabulary Terms and Phrases - Continued

Term Or Phrase	Documents	Document Prevalence	Mean Tf-Idf	Phrase Length	Vocabulary Score
shoulder	65	3.2%	0.0073	1	0.0073
walking	66	3.3%	0.0072	1	0.0072
fatal	65	3.2%	0.0072	1	0.0072
high	86	4.3%	0.0072	1	0.0072
rib	65	3.2%	0.0072	1	0.0072
lifting	56	2.8%	0.0071	1	0.0071
ankle	62	3.1%	0.0071	1	0.0071
drill	56	2.8%	0.0070	1	0.0070
bucket	61	3.0%	0.0070	1	0.0070
end	72	3.6%	0.0069	1	0.0069

Conclusions and Recommendations

1. The updated preprocessing reduces the influence of common English words and generic employee references.

2. Nine clusters provide a simpler exploratory structure than the earlier fifteen-cluster solution while retaining a nearly equivalent overall silhouette score.

3. The clusters should be interpreted through top terms, representative narratives, post-hoc MSHA fields, and severity distributions.

4. 2 clusters had a mean silhouette value at or below zero. Those clusters should be considered mixed or weakly defined.

5. 531 individual narratives had negative silhouette values. These narratives may contain multiple topics or fit another cluster nearly as well.

6. Clear groups such as hearing loss, roof falls, truck incidents, hand/finger injuries, and ankle-twisting incidents provide evidence that narrative text contains discoverable recurring patterns.

7. Clusters based primarily on injury outcome or reporting language may be less useful than clusters organized around event mechanisms, hazards, equipment, or work activities.

8. Existing MSHA categories were not used to create the clusters. The post-hoc comparison should be used to identify clusters that connect records across several official classifications.

9. This remains a supporting proof of concept. The main final-project direction remains the extraction of reported causes, contributing factors, and preliminary corrective-action categories.

Technical Appendix

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Input narratives: 2,000
TF-IDF vocabulary size: 2,763
Minimum document frequency: 3
Maximum document frequency: 70%
N-gram range: (1, 2)
Maximum vocabulary size: 8,000
SVD components: 75
SVD explained variance: 0.2270
Selected cluster count: 9
Selected silhouette score: 0.0533
Highest tested score: 0.0736
Highest-scoring cluster count: 15
Random state: 42

PDF output: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final
Project/MSHA_Narrative_Clustering_Proof_of_Concept_Updated.pdf
Cluster assignment CSV: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW
Data Science/MSDS 453- AI and Natural Language/Final
Project/MSHA_Narrative_Cluster_Assignments_Updated.csv
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